Yutian Zhang

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EDUCATION

The Hong Kong University of Science and Technology 2024 - Present

Ph.D. in Electronic and Computer Engineering | Advisor: Prof. Mansun Chan

- Research focus: device-circuit DTCO of CFET SRAM under practical layout, stability, and parasitic constraints.

The Hong Kong University of Science and Technology 2020 - 2024

B.Eng. in Electronic and Computer Engineering

- GPA: 3.62/4.0; Major GPA: 3.91/4.0; Rank: 3/133 (top 2%).

PUBLICATIONS

[1] Y. Zhang, K. Sarfraz, and M. Chan, *"DTCO-based Hybrid Rail 8T Complementary FET SRAM Design towards Advanced Node,"* Accepted.

[2] Y. Zhang, K. Sarfraz, and M. Chan, *"DTCO-based Monolithic Vertical Four-layer 6T Complementary FET SRAM Design,"* Accepted.

[3] Y. Zhang, J. Fang, K. Sarfraz, and M. Chan, *"CFET SRAM Design Manuscript,"* Manuscript under review.

RESEARCH EXPERIENCE

Device-Circuit DTCO of Complementary FET SRAM 2024 - Present

Ph.D. Researcher, HKUST | Advisor: Prof. Mansun Chan

- Study CFET as a circuit-technology design space where device architecture, layout, tier configuration, parasitics, and circuit metrics are co-optimized.
- Develop SRAM-centered DTCO studies connecting vertical CFET integration to density, access performance, stability, energy, and array-level PPA outcomes.
- Investigate CFET SRAM layout and sizing options, including cell organization and read/write stability tradeoffs, with detailed concepts disclosed after publication.

SRAM and DRAM Based In-Memory Computing Feb 2023 - 2024

Undergraduate Research Assistant, HKUST | Advisor: Prof. Shao Qiming

- Designed 6T SRAM for binary neural-network operation using split word lines, reducing average word-line current by approximately 30%.
- Designed a small-offset voltage-mode sense amplifier and evaluated write-noise-margin improvement through Monte Carlo analysis.

Analog Circuit Design Internship Jul 2023 - Aug 2023

InstonTech, Suzhou, China

- Analyzed conventional 6T SRAM behavior and proposed a 9T SRAM design with improved static-noise-margin characteristics.
- Reduced write power through an improved latch-based precharge circuit and supported module-level reverse engineering from netlist information.

SELECTED PROJECTS

Hybrid Rail 8T CFET SRAM; Monolithic Four-layer 6T CFET SRAM; Ongoing CFET SRAM DTCO Study

- Accepted and under-review CFET SRAM works covering hybrid rail design, monolithic vertical four-layer SRAM, and broader device-circuit DTCO questions.
- Future Ph.D. modules will be disclosed after the corresponding work is published.

TECHNICAL SKILLS

Device / circuit	CFET SRAM design, DTCO, SRAM stability analysis, PPA evaluation
EDA / simulation	Cadence Virtuoso, HSPICE/Spectre, Silvaco, VHDL
Programming	Python, MATLAB, C/C++

HONORS

Dean's List and Academic Achievement Award

The Hong Kong University of Science and Technology